2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), 13-15 February 2025, CUET, Chattogram-4349, Bangladesh

2025 International Conference on Electrical, Computer and Communication Engineering (ECCE) | 979-8-3503-5750-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ECCE64574.2025.11013133

An Attention-based Hybrid Deep Learning Approach for Skin Lesion Classification

Saiba Sufia[1], Md. Farhad Hossain[2], and Md. Azad Hossain [3]
[1,2,3]*Department of Electronics and Telecommunication Engineering*
Chittagong University of Engineering and Technology, Chattogram 4349, Bangladesh
Email - saiba.sufia1212@gmail.com, farhad.hossain@cuet.ac.bd, azad@cuet.ac.bd

***Abstract*—Skin cancer is the most frequent type of cancer worldwide in recent years. It affects more than a million people annually, a large number of whom are elderly. Because of incredibly fine-grained differences of skin lesion, it is difficult to classify distinct forms of skin lesions using images. An automated system ought to be able to classify these skin lesions with reasonable accuracy. Preprocessing steps like Blackhat filter, Median filter including resizing, augmentation and labeling were performed before uploading the images to the system. For the model, a sizable number of images must be used for training to yield the required results. This model is trained on 2 different dataset: HAM10000 (Human Against Machine) with 10,015 images on skin lesion and ISIC(The International Skin Imaging Collaboration). Here A hybrid CNN model combined with soft attention was developed for effective multi-class classification. Here, ConvNet (A customized CNN) and DenseNet-201 is used to the proposed hybrid model. In previous several models were implemented and it is found that, by combining the best characteristics of both architectures, the DenseNet-ConvNet model has the potential to improve feature learning by fusioning low level and high level features, increase accuracy, and even increase parameter efficiency. The model yields an accuracy of 96.13% for HAM-10,000 and 92.36% for ISIC dataset, attaining f1-score of 96% on HAM-10,000 and 92.37% on ISIC.**

***Index Terms*—Skin lesion, ConvNet, DenseNet, Soft-attention, Hybrid, ISIC, Fusion**

I. Introduction

The 19 th most prevalent malignancy in both men and women is melanoma of the skin [1]. Melanoma in particular is a complicated form of skin cancer. Only 1% of all skin cancers are malignant melanoma, but it causes the vast majority of skin cancer fatalities. Europe, North America, and Oceania are the regions that are most impacted. Since the middle of the 1970s, the incidence of invasive melanoma has been steadily rising. The rate climbed by around 2% year between 2008 and 2017 [2]. The American Cancer Organization estimates that 106,110 new cases of cutaneous melanoma were diagnosed in the U.S. in 2021, while 7,180 people also passed away from the condition. Europe, the United States, Canada, and Australia are the continents where skin melanoma occurs most frequently worldwide. Although cutaneous melanoma has a high 5-year survival rate (93%), early disease identification is crucial to lower melanoma-related mortality [3]. Skin cancer is a form of cancer when a lesion first appears on the skin and then progresses to cancer. Early skin cancer detection is crucial since it can be challenging to discern between a normal mole and a dangerous lesion, making it impossible to find the cancerous cells at an early stage. Skin cancer cases are growing every day. According to records, there were about 76,250 new cases of melanoma and 8,790 new melanoma-related deaths in the US in 2012. Risk Factors for Melanoma Skin Cancer lists numerous variables that may have an impact on each individual's risk. Melanoma's tendency to spread is mostly caused by sun exposure, a lack of early identification, a high life expectancy of the population, etc.

Deep Learning techniques use several processing layers to develop a hierarchical understanding of the data. They offer a way to utilize enormous volumes of data while requiring little manual feature engineering. Beginning with the release of AlexNet in 2012, Deep Learning has seen impressive breakthroughs in the field of Computer Vision [4].

This study's main goal is to investigate how well visual information can be classified as skin lesions. Furthermore, in order to improve classification accuracy, the study evaluates the effectiveness of a brand-new hybrid technique.

The main contributions of the stated methodology are what follows:

1) The dataset is preprocessed to remove hair and artifacts. This study uses a hybrid network of ConvNet and Densenet-201 for the extraction of feature effectively including edges, forms and textures.
2) Improving results by using soft attention that fuses features and focuses on relevant portions. Can classify most of the skin lesions correctly with high accuracy in both balanced and imbalanced dataset .

This paper explains our proposal in a structured way. Section 2 focuses on the literature review with research gap , Section 3 gives a description about dataset , Section 4 presents the methodology of this skin lesion analysis, Section 5 depicted result analysis and Section 6 discussed with the conclusion and future work.

II. Literature Review

A. Related Work

Dermatology has undergone a revolution with the development of machine learning-based skin lesion identification, which has replaced laborious feature extraction techniques with complex deep learning methods.

Early approaches to skin lesion categorization mostly relied on feature-based techniques like Support Vector Machines (SVM), k-nearest-neighbor algorithm (k-NN) [6] and Random Forest [5]. A flexible ensemble learning method called Random Forest [5] builds several decision trees throughout training and combines their predictions to improve classification accuracy.

Convolutional neural networks, or CNN [7] are a major advancement in image processing, especially for applications like pattern recognition and picture classification. Without the requirement for explicit feature engineering, CNNs automatically recognize hierarchical properties by filtering incoming data mostly images through a number of layers. Local patterns such as edges and textures are analyzed by these networks using convolutional layers [7], and the subsequent layers combine data to obtain increasingly abstract representations with less spatial dimensions. Usually, the architecture ends with fully connected layers that handle tasks like classification or regression.

Using large datasets such as ImageNet [8], transfer learning - a potent deep learning technique has transformed dermatology by utilizing pre-trained Convolutional Neural Networks, models. When used with dermatology datasets, these pre-trained models which were first trained on extensive general image recognition tasks offer a notable benefit, especially in tasks involving the classification of skin lesions. Numerous well-known architectures have been thoroughly examined, improved, and tailored for this particular area, notably VGG [9], DenseNet [10], Inception [11] and ResNet [12].

Ensemble machine learning techniques combine multiple models to enhance prediction accuracy [11]. This enhances robustness to noisy input, lessens overfitting, and improves generalization. They are particularly useful in ensuring stability, fostering adaptation, managing complex interactions, and eliminating bias by accommodating a range of base learners [18].

B. *Research Gap*

Despite the progress in skin lesion detection using machine learning, some approaches still need improvement. Masato Tada and Xian-Hua Han [13] suggested a hybrid skin lesion classification model which can be improved by further analysis of different attention mechanism. In another work, authors [14] finds that, before classifying, Effective techniques of preprocessing and model improving can enhance detection performance.

For classification problems, using ensemble methods may lead to inaccurate assessment of the importance of features derived from each model. Stacking, on the other hand, entails passing the image across models one after the other, suggesting that each model after that may change how relevant the features that were retrieved are. In light of these drawbacks, this research suggests a hybrid feature extracting strategy that combines conventional pre-trained CNN models like DenseNet-201 and ConvNet. In contrast to ensembling, which combines features from many models after classification, our design makes sure that every model extracts features on its own before fusion. Because each model's features remain relevant, this approach improves performance.

III. Dataset

HAM10000 (Human Against Machine with 10,000 Training Images) was used in this study and it helps to develop skin lesion categorization [15] . There are 10,015 dermatoscopic images in the dataset. Clinical details, including patient age, sex, and the anatomical site of the lesion, are linked to each image in the dataset. A HAM10000 image had dimensions of 450 pixels by 600 pixels and had the three RGB (Red, Green, and Blue) color channels, which are typical for color images, for a total of 810K numbers per image. Typically, these datasets are kept in popular image file types like JPEG or PNG. The International Skin Imaging Collaboration (ISIC) provided 2357 images in this set [22].

TABLE I: Number of images in different classes

Skin Pathology	ISIC	HAM-10000
Actinic keratosis	130	327
Basal cellcarcinoma	392	514
Dermatofibroma	111	115
Melonoma	454	1113
Nervus	373	6705
Pigmented benign keratosis	478	1099
Vascular lesion	142	142
seborrheic keratosis	80	-
squamous cell carcinoma	197	-

IV. Proposed Methodology

The proposed design was developed using two distinct architectural styles, and after that their features are fused and attention technique was used. The model was based on pretrained model and a customized convolutional neural network.

We utilize the most advanced pre-trained convolutional neural network designs, such as Dense-Net-201 and ConvNet, for the extraction of visual features. These designs are suitable for extracting significant visual information from images of skin lesions and have shown outstanding performance in image identification tests. The DenseNet-ConvNet hybrid connection shown in Fig.1 combines conventional Convolutional Neural Network (ConvNet) architectures with the DenseNet design, which is renowned for its dense connection patterns that promote feature reuse. Each layer may have direct access to the feature map of all layers before it because to this design, which facilitates effective propagation of features throughout the network. In addition to improving gradient flow and making training easier, this extensive connection encourages feature reuse, which helps the model make efficient use of data from previous layers. The hybrid DenseNet-ConvNet model with Soft-attention outperforms other computer vision models by fusing the best features of both architectures.

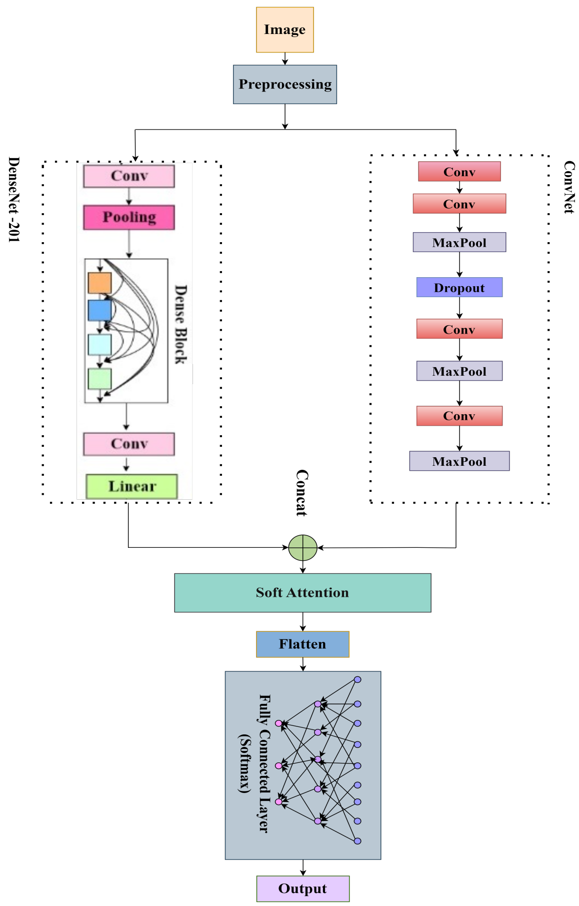

Fig. 1: Proposed Model Architecture

A. *Image Preprocessing*

Image preprocessing methods, particularly the Blackhat and Median filters, were applied initially. The Median filter is used to cut down on noise, and the Blackhat filter is intended to draw attention to dark details. Label Encoding, a method for transforming categorical data into numerical format for machine learning models, is also introduced in the article. In order to increase the data in ISIC, data is augmented (rotation, width shift, height shift, horizontal flip, vertical flip) to increase the number of images to 1000 for every class.

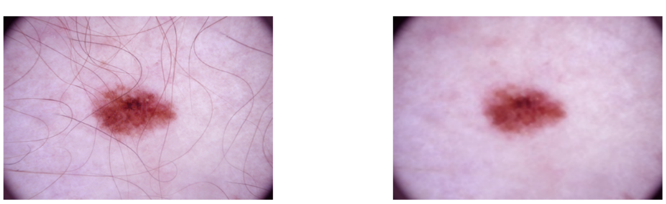

Fig. 2: Image before and after preprocessing.

B. *Feature Extraction With Separable Convolution*

Separable convolution is used here in image processing to lower computational complexity while maintaining performance. Using this method, a 2D convolution is divided into two one-dimension convolutions-horizontal and vertical axis, maintaining the fundamental characteristics of conventional 2D convolution. This method is very helpful in contexts with limited resources because it drastically lowers the amount of parameters and computations needed.

TABLE II: ConvNet Architecture .

Layer	Output Shape	Parameter
input	(None, 224, 224, 3)	0
seperable conv2d(Conv2D)	(None, 224, 224, 256)	1051
seperable conv2d(Conv2D)	(None, 224, 224, 198)	53190
max_pooling2d(MaxPooling 2D)	(None, 56, 56, 198)	0
dropout(Dropout)	(None, 56, 56, 198)	0
seperable conv2d(Conv2D)	(None, 56, 56, 64)	14518
max_pooling2d(MaxPooling 2D)	(None, 14, 14, 64)	0
seperable conv2d(Conv2D)	(None, 14, 14, 32)	2656
max_pooling2d(MaxPooling 2D)	(None, 7, 7, 32)	0
Total parameter: 71415 (278.96 KB)		

C. *Feature Extraction With Densenet-201*

DenseNet-201 is a deep convolutional neural network architecture that stands out for having densely coupled layers, meaning that every layer receives inputs from every level that came before it. This architecture's effective use of features throughout the network results in strong gradient flow and improved feature reuse. With 201 layers, it's dense connections is very useful at capturing minute details and variances in images, which may be essential for distinguishing between various kinds of skin diseases.

D. *Proposed Hybrid Based Framework*

DenseNet-201 is used here for their effective feature reuse, but Convolutional Neural Networks (CNNs) are excellent at capturing spatial data. There are various benefits to combining multiple architectures into a hybrid approach. Through the merging of information from many layers and the mitigation of the vanishing gradient issue, this hybrid technique in Fig.3 improves performance across a range of image processing tasks. Integrating data from several sources or layers is made possible by spatial domain concatenation, a crucial method in this context that improves the model's capacity to collect high-level and low-level information. In the end, this produces models that are more reliable and accurate.

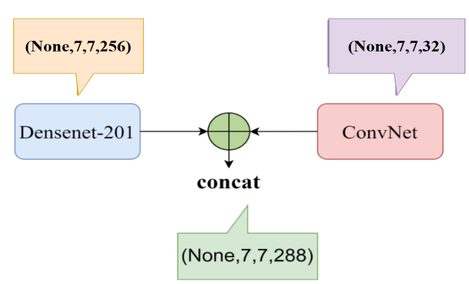

Fig. 3: Hybrid connection between ConvNet and DenseNet-201.

E. Soft-Attention

The attention block uses a 3D convolution to extract information. Next, weights emphasizing important areas are calculated using a combination of the multiplication and softmax algorithm. Prior to merging these weights with an adaptable bias for additional refinement, they are utilized to highlight pertinent features that were previously extracted. This enables the model to concentrate on important image details that could be necessary for correctly classifying skin lesions [16].

$$f_{sa} = \gamma t \left(\sum_{k=1}^{K} \text{softmax}(W_k * t) \right) \tag{1}$$

The feature tensor in 1 $t \in \mathbb{R}^{h \times w \times d}$is the input for a 3D convolutional layer , where the the weights $W_k \in \mathbb{R}^{h \times w \times d \times K}$, with K, where K is the number of 3D weights. The softmax function is used to normalize the result of this convolution, resulting in K=16 attention maps.

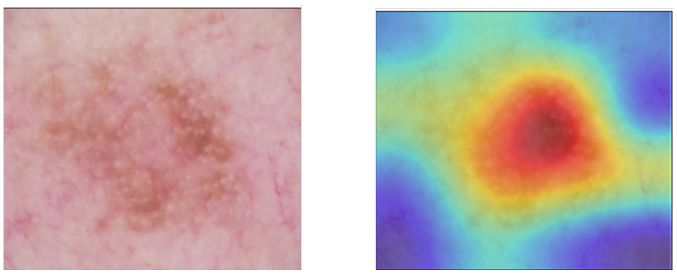

Fig. 4: Heatmap after passing image through Hybrid model with soft attention

F. Choice of Hyperparameters

Hyperparameters shown in Table.III help neural networks be trained efficiently, encourage convergence, avoid overfitting, and perform well in classification tasks.

TABLE III: Hperparameter Values

Hyperparametrs	**Value**
Loss function	Sparse categorical crossentropy
Epochs	40
Dropout	0.5
Optimizer	Adam
Learning rate	4×10^{-4}
Batch size	4

Training results can be further optimized in Table IV by modifying these parameters in accordance with the features of the dataset and the complexity of the model.

TABLE IV: Performance Comparison of Different Optimizer

Optimizer	**Accuracy**	**Recall**	**precision**
RmsProp	81.27	81.27	77.87
Adam	96.13	96.13	96.12

It is observed in Fig.5 that when the learning rate is increased gradually 10 times from 0.0004, the accuracy becomes decreased. Also, the model is unable to classify all the classes equally as they results of low precision and recall.

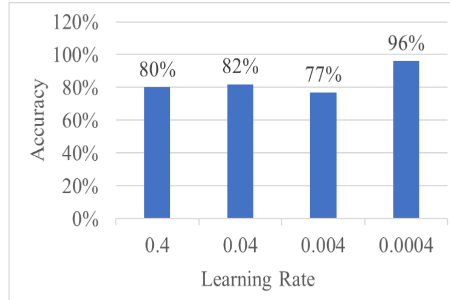

Fig. 5: Effects of different learning rate on accuracy.

V. Result Analysis

We have developed a hybrid model by combining ConvNet and DenseNet-201 with attention in which the "HAM-10000" dataset, the suggested approach (Hybrid model) obtained the best weighted accuracy of 96%, in contrast to the accuracy of 93.4% of the baseline approach (i.e., Soumyya et al. [16]). In a similar vein, the suggested model achieved the greatest accuracy of 92% for the 'ISIC-2019" dataset when compared to the results of the Soumyya et al. model [16]. The comparison's analysis proved that, for both datasets, the suggested method performed better than other recent efforts.

A. Performance Comparison of CNN Models

The Table V clearly shows a trend overall: ensemble approaches and the incorporation of attention mechanisms greatly improved baseline models' performance. Among the skin lesion classification methods tested on this dataset, the suggested DenseNet-201 Hybrid model performs the best in terms of accuracy, recall, and precision.

TABLE V: Performance Comparison of Different Models

Models	**Accuracy**	**Recall**	**Precision**
Inception V3	70.23	70.23	68.77
ConvNet	76.46	76.28	75.24
DenseNet-201	78.34	78.30	77.56
ConvNet + Hard-attention	80.12	80.12	64.41
ConvNet + Multihead-attention	86.71	86.71	86.42
Vgg-16 + Soft-attention	87.25	87.25	85.40
ConvNet + Soft-attention	93.96	93.96	93.85
ResNet-50 + Soft-attention	94.44	94.44	94.87
DenseNet-201 + Soft-attention	94.92	94.92	93.34
ResNet-50 + CNN ensemble	94.42	94.32	94.36
DenseNet-201 + CNN ensemble	95.28	95.28	94.36
ResNet-50 + CNN + Soft-attention	95.21	95.21	92.84
Our proposed approach	**96.13** **92.36**	**96.13** **92.36**	**96.12 (HAM)** **92.58 (ISIC)**

B. Results

Here, precision, recall, F1-score was comapared between each class of HAM-10,000 and ISIC in Table VI and VII. Also models for multiclass classification are assessed using Receiver Operating Characteristic (ROC) curves in Fig.7. The percentage of positive instances that were accurately identified as positive is known as the TPR.

TABLE VI: Comparison of Different Classes(HAM-10,000).

Classes	Precision	Recall	F1-score	AUC
Actinic keratosis	0.83	0.83	0.83	0.910
Basal cell carcinoma	0.95	0.81	0.88	0.903
Dermatofibroma	0.85	0.94	0.89	0.916
Melonoma	1.00	0.83	0.91	0.821
Nervus	0.88	0.65	0.75	0.956
Pigmented benign keratosis	0.98	0.99	0.99	0.962
Vascular lesion	1.00	0.90	0.95	0.950

TABLE VII: Comparison of Different classes(ISIC).

Classes	Precision	Recall	F1-score	AUC
Actinic keratosis	0.97	0.95	0.96	0.967
Basal cell carcinoma	0.87	0.93	0.90	0.922
Dermatofibroma	0.94	0.98	0.96	0.985
Melonoma	0.87	0.86	0.87	0.970
Nervus	0.89	0.95	0.92	0.920
Pigmented benign keratosis	0.94	0.85	0.89	0.957
Seborrheic keratosis	0.98	0.90	0.94	0.952
Vascular lesion	1.00	0.99	0.99	0.947
Squamous cell carcinoma	0.84	0.93	0.88	0.994

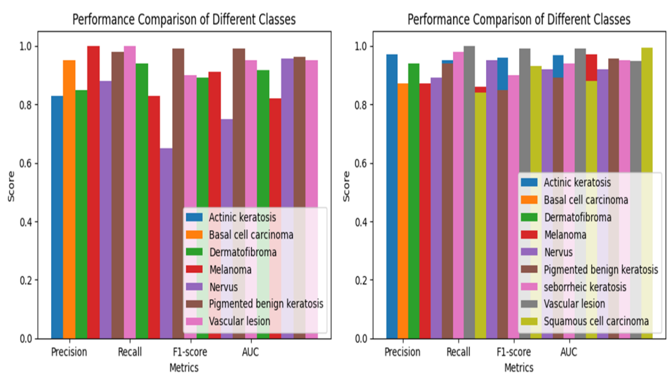

Fig. 6: Comparison between different classes of HAM-10,000 (left) and ISIC (right).

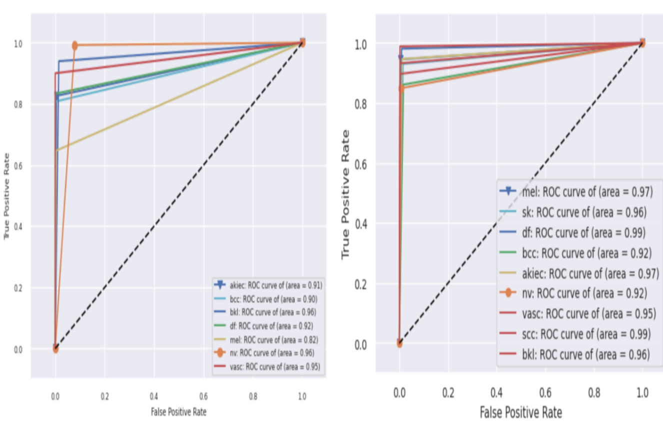

Fig. 7: Receiver Operating Characteristic Curve of HAM-10,000(left) and ISIC (right)

C. Error Analysis

Two types of error analysis are discussed here which are quantitative and quanlitative analysis.

1) *Quantitative Analysis:*

Confusion matrix in Fig.8 is used to assess how well an image categorization model performs. Here, it is visualised that this model can identify correctly most of the images on both balanced and unbalanced dataset.

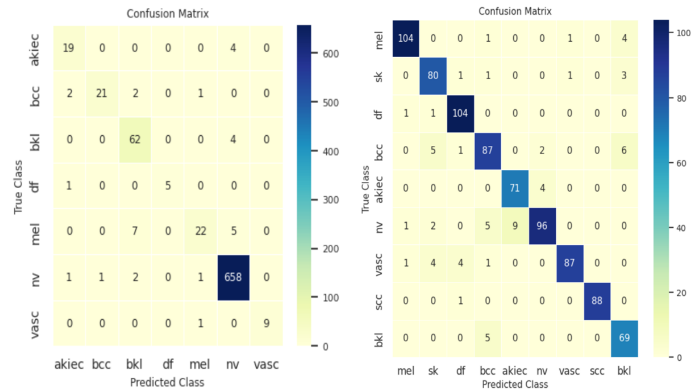

Fig. 8: Confusion Matrix of HAM-10,000(left) and ISIC(right).

2) *Qualitative Analysis:*

Skin lesions can vary greatly in appearance from one another, and identifying them can be challenging due to the cluttered and unclear surroundings surrounding the lesion. It is difficult to determine the type of lesion because of these two variables. but , our model can correctly identify most of the skin lesions shown in Fig.9.

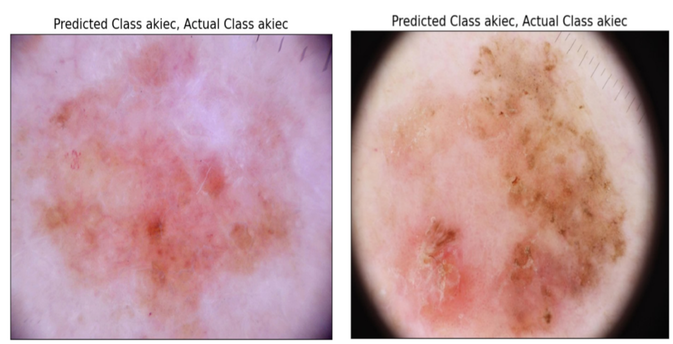

Fig. 9: Example of correct classification using hybrid- attention model (HAM-10,000).

Sometimes, because of the poor image quality, the algorithm may have had trouble correctly classifying the images. As a result, the system incorrectly identified the image as an actinic keratosis and melanoma.

D. Comparison With Existing Works

Here, Table VIII demonstrates that, in comparison to all the recent approaches, the suggested hybrid model attains more accuracy on the HAM-10000 and ISIC datasets. This implies that the suggested method represents a more effective strategy for classifying skin lesions.

TABLE VIII: Comparative Analysis of The Proposed Method With The Existing Models.

Techniques	Datasets	Accuracy (%)
Rajit et al. [17]	HAM-10000	81
Danish et al. [21]	ISIC -2019	86.47
Masato et al. [13]	HAM-10000	91
	ISIC -2019	87
S.M. Mahedy et al. [18]	HAM-10000	94
Soumyya et al. [16]	ISIC-2017	90
	HAM-10000	93.4
Zillur et al. [19]	ISIC-2017	93
	HAM-10000	94
Mohamed et al. [20]	ISIC-2019	94.92
S.p.Godlin et al. [14]	HAM-10000	95
Proposed (Hybrid + Soft-attention)	**HAM-10000**	**96.12**
	ISIC	**92.36**

VI. Conclusion and Future Work

In dermatology, multi-class skin lesion detection is essential for precise diagnosis and suitable therapy. The goal of this work is to use the HAM10K and ISIC dataset to increase the skin lesion recognition accuracy. To address the difficulties presented by this dataset, a number of pre-processing methods, such as BlackHat filtering, Digital hair removal (DHR), median filtering and augmentation(for ISIC) is used. This illustrates the potential and efficacy of the Soft Attention-based deep learning framework in image processing. The Soft Attention function enhances the main network's performance by internally providing the location of the model's focus during disease classification. In order to improve classification accuracy, a hybrid approach of Convolutional Neural Networks (CNNs) in the classification of skin lesions combines a variety of architectural elements and methodologies. Finally the accuracy of 96% was achieved through the proposed model. By addressing the practical challenges in health care and diagnosis, updating the dataset, refining the model, and completing an additional evaluation for more classes will all contribute to better results in future work. Additionally, Transformers could be implemented for improved classification.

References

[1] *Skin Cancer Statistics Melanoma of the Skin is the 19th Most Common Cancer Worldwide.*, *American Institute for Cancer Research.*, no. 8, 2022. Springer. Available at: https://www.mdpi.com/2075-4418/11/8/1390.

[2] *American Cancer Society*, *Cancer Facts and Figures*, no. 8, 2021. Available at:https://www.cancer.org/content/dam/cancer-org/research/cancer-factsand-statistics/annual-cancer-facts-and-figures/2021/cancer-facts-andfigures-2021.pdf.

[3] Nie, Yali and Sommella, P. and Carratù, Marco and Ferro, Matteo and O'Nils, Mattias and Lundgren, Jan, *Recent Advances in Diagnosis of Skin Lesions Using Dermoscopic Images Based on Deep Learning*, *IEEE Access*, 2022, pp. 1-1. IEEE Access. [Online; accessed January-2022]. Available at: https://doi.org/10.1109/ACCESS.2022.3199613.

[4] Hosny, Khalid M. and Kassem, Mohamed A. and Foaud, Mohamed M., *Classification of skin lesions using transfer learning and augmentation with Alex-net*, *PLOS ONE*, 2019, 14. Available at: https://doi.org/10.1371/journal.pone.0217293.

[5] Raju Gottumukkala, V. S. S. P. and Kumaran, N. and Chandra Sekhar, V., *IRFNet: Skin Lesion Detection and Classification Using Unified Intuitive and Object Classifier with Iterative Random Forest Algorithm*, *International Journal of Image and Graphics*, Available at: https://doi.org/10.1142/S021946782340003X.

[6] Ballerini L., Fisher R., Aldridge B., and Rees J., *A color and texture based hierarchical K-NN approach to the classification of non-melanoma skin lesions*, *Springer*, 2013,vol. 6, pp. 63-86. Available at: https://doi.org/10.1007/978-94-007-5389-1_4.

[7] Yu, Chanki and Yang, Sejung and Kim, Wonoh and Jung, Jinwoong and Chung, Kee-Yang and Lee, Sangwook and Oh, Byungho., *Correction: Acral melanoma detection using a convolutional neural network for dermoscopy images*, *PLOS ONE*, 2018, vol. 13, Available at: https://doi.org/10.1371/journal.pone.0196621.

[8] Krizhevsky, Alex and Sutskever, Ilya and Hinton, Geoffrey, *ImageNet Classification with Deep Convolutional Neural Networks*, *Neural Information Processing Systems*, 2012, vol. 25, Available at: https://doi.org/10.1145/3065386.

[9] Naronglerdrit, Prasitthichai and Mporas, Iosif, *Evaluation of Big Data based CNN Models in Classification of Skin Lesions with Melanoma*, *Neural Information Processing Systems*, 2020, vol. 25, Available at: https://doi.org/10.1007/978-981-15-6321-8_5.

[10] Adegun, Adekanmi and Viriri, Serestina, *FCN-Based DenseNet Framework for Automated Detection and Classification of Skin Lesions in Dermoscopy Images*, *IEEE Access*, 2020, pp 1-1, Available at: https://doi.org/10.1109/ACCESS.2020.301665.

[11] Nils Gessert and Maximilian Nielsen and Mohsin Shaikh and René Werner and Alexander Schlaefer, *ImageNet Classification with Deep Convolutional Neural Networks*, *MethodsX*, 2020, vol. 7, Available at: https://www.sciencedirect.com/science/article/pii/S2215016120300832.

[12] Kaiming He and Xiangyu Zhang and Shaoqing Ren and Jian Sun, *Deep Residual Learning for Image Recognition*, *arXiv*, 2015, Available at: https://doi.org/10.48550/arXiv.1512.03385.

[13] Tada, Masato and Han, Xian-Hua, *Bottleneck Transformer model with Channel Self-Attention for skin lesion classification*, *2023 18th International Conference on Machine Vision and Applications (MVA)*, 2023, vol. 7,pp.1-5 Available at: https://doi.org/10.23919/MVA57639.2023.10215720.

[14] Jasil, S. and Venugopal, Ulagamuthalvi, *A hybrid CNN architecture for skin lesion classification using deep learning*, *Soft Computing*, 2023, vol. 7,pp.1-10 Available at: https://doi.org/10.1007/s00500-023-08035-w.

[15] Philipp Tschandl and Cliff Rosendahl and Harald Kittler, *The HAM10000 Dataset: A Large Collection of Multi-Source Dermatoscopic Images of Common Pigmented Skin Lesions*, *arXiv*, 2018, vol. 7,pp.1-5 Available at: https://dblp.org/rec/journals/corr/abs-1803-10417.bib.

[16] Datta, Soumyya Kanti and Hashemi, Mohammad Abuzar and Srihari, Sargur and Gao, Mingchen, *Soft-Attention Improves Skin Cancer Classification Performance*, , 2021, Available at: https://doi.org/10.1101/2021.05.12.21257114.

[17] Hajiarbabi, Mohammadreza and Chandra, Rajit, *Skin Lesion Detection Using Deep Learning*, *Journal of Automation, Mobile Robotics and Intelligent Systems*, 2023, vol. 7,pp.56-64 Available at: https://doi.org/10.14313/JAMRIS/3-2022/24.

[18] Hasan, S M and Mamun, Md. Al and Faruk, Md and Srizon, Azmain *Ensemble of Deep Convolutional Neural Networks for Multi-Class Skin Lesion Recognition Using Soft Attention*, 2023,pp.65-69 Available at: https://doi.org/10.1109/ICICT4SD59951.2023.10303518.

[19] Zillur Rahman and Md. Sabir Hossain and Md. Rabiul Islam and Md. Mynul Hasan and Rubaiyat Alim Hridhee *An approach for multiclass skin lesion classification based on ensemble learning*, *Informatics in Medicine Unlocked*, 2021, vol. 25 ,pp.65-69 Available at: https://doi.org/10.1016/j.imu.2021.100659.

[20] Kassem, Mohamed and Hosny, Khalid and Fouad, M. *Skin Lesions Classification Into Eight Classes for ISIC 2019 Using Deep Convolutional Neural Network and Transfer Learning*, *IEEE Access*, 2020, vol. 25 ,pp.1-1 Available at: https://doi.org/10.1109/ACCESS.2020.3003890.

[21] Jamil, Danish and Qazi, Farheen and Agha, dur-e-shawar and Palaniappan, Sellappan *Classification of skin lesion using deep convolutional neural network by applying transfer learning*, *Journal of Autonomous Intelligence*, 2023, vol. 6 , Available at: https://doi.org/10.32629/jai.v6i3.747.

[22] ANDREY KATANSKIY *ISIC Dataset*, 2018 , Available at: https://www.kaggle.com/datasets/nodoubttome/skin-cancer9-classesisic.